



Computer  
Science

# CSC380: Principles of Data Science

**Probability 3**

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# Review: “probability cheatsheet”

2

## Additivity:

For any *finite* or *countably infinite* sequence of disjoint events  $E_1, E_2, E_3, \dots$ , 
$$P\left(\bigcup_{i \geq 1} E_i\right) = \sum_{i \geq 1} P(E_i)$$

## Inclusion-exclusion rule:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Law of total probability: For events  $B_1, B_2, \dots$  that partitions  $\Omega$ ,

$$P(A) = \sum_i P(A \cap B_i)$$

## Conditional probability:

$$P(A|B) := \frac{P(A \cap B)}{P(B)}$$

$(P(A|B) \neq P(B|A) \text{ in general})$

## Probability chain rule:

$$P(A \cap B \cap C) = P(A|B \cap C)P(B|C)P(C)$$

Law of total probability + Conditional probability:

$$P(A) = \sum_i P(A \cap B_i) = \sum_i P(B_i)P(A|B_i) = \sum_i P(A)P(B_i|A)$$

## Bayes' rule:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B|A)P(A)}{P(B)}$$

## Independence:

(definition) A and B are independent if  $P(A, B) = P(A)P(B)$

(property) A and B are independent if and only if  $P(A|B) = P(A)$  (or  $P(B|A) = P(B)$ )

- Random variables
- Distribution functions
  - probability mass functions (PMF)
  - cumulative distribution function (CDF)

# Random Variables

# Random variables (RVs)

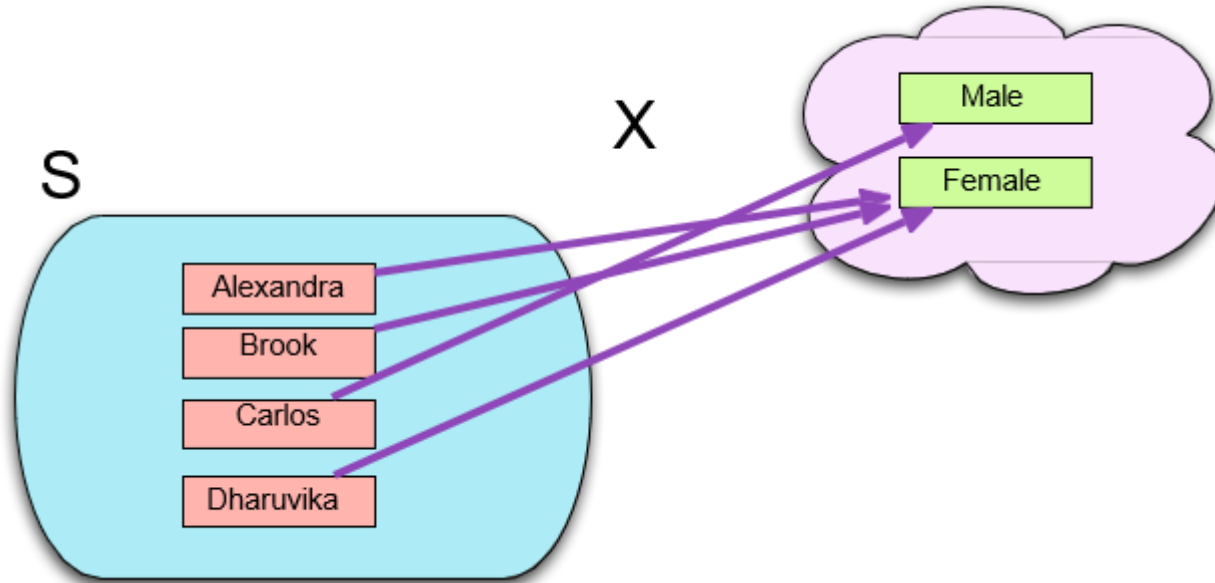
- A single random sample may have more than one characteristic that we can observe (i.e., it may be bi-/multivariate data).
- We can represent each characteristic (e.g., gender, weight, cancer status, etc.) using a separate random variable.

## Random Variable

A **random variable** connects each possible outcome in the sample space to some property of interest.

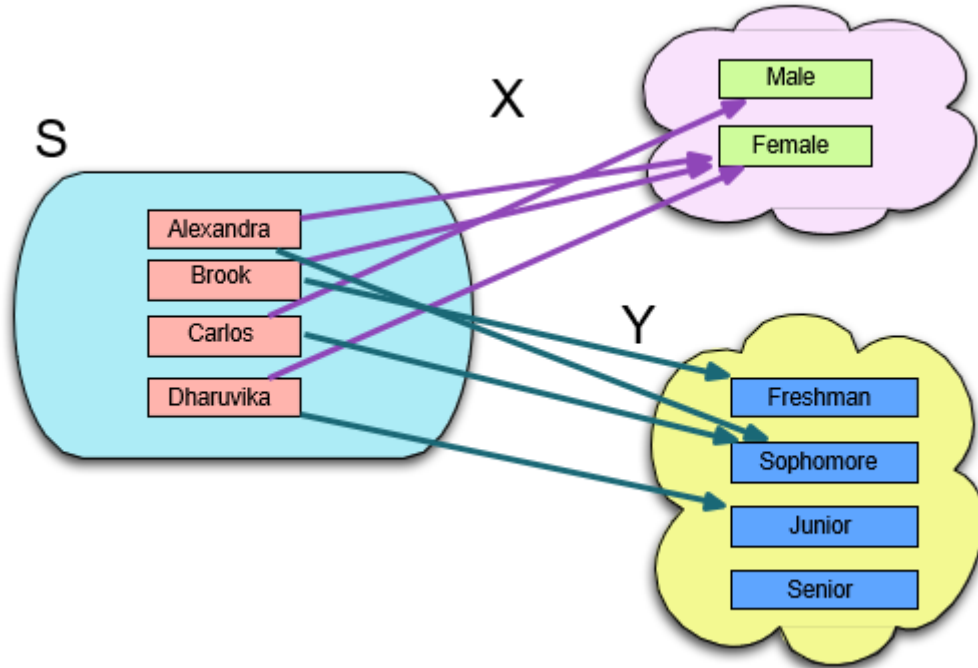
Each value of the random variable (e.g., male or female) has an associated probability.

# Random Variable: Example



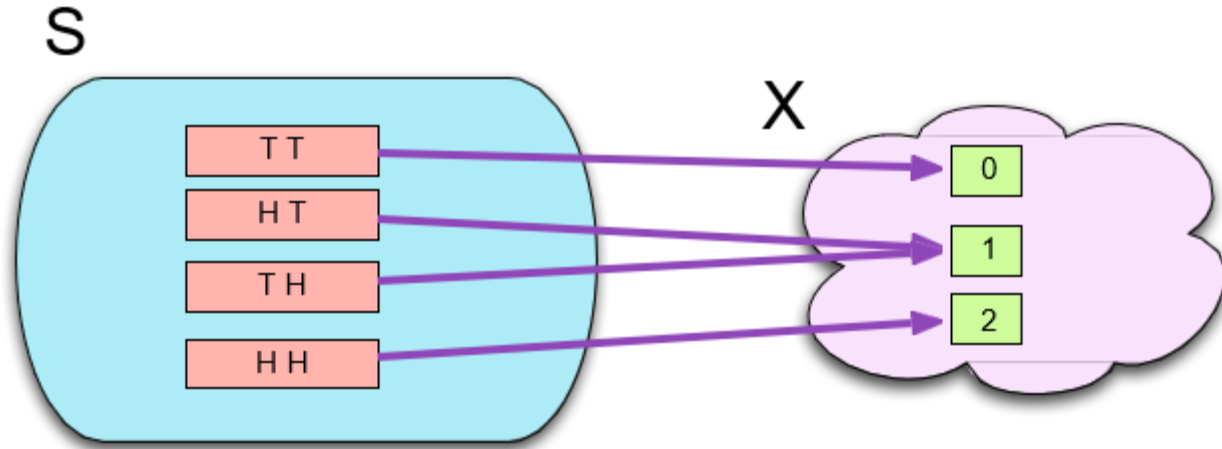
- $X$ : people  $\rightarrow$  their genders

# Random Variable: Example



- $Y$ : people  $\rightarrow$  their class year

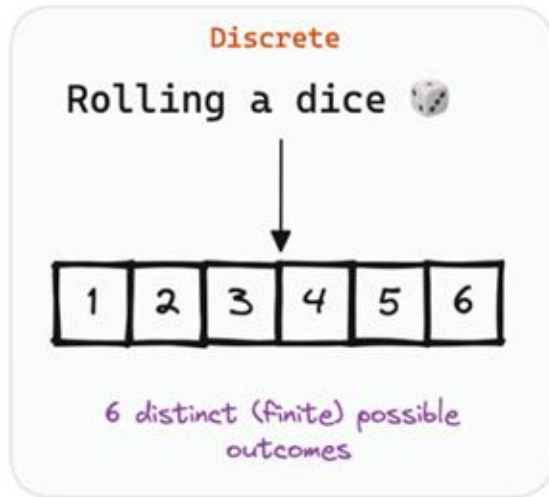
# Random Variable: Example



- $X$ : sequence of coin flips  $\rightarrow$  Number of heads

# Types of Random Variables

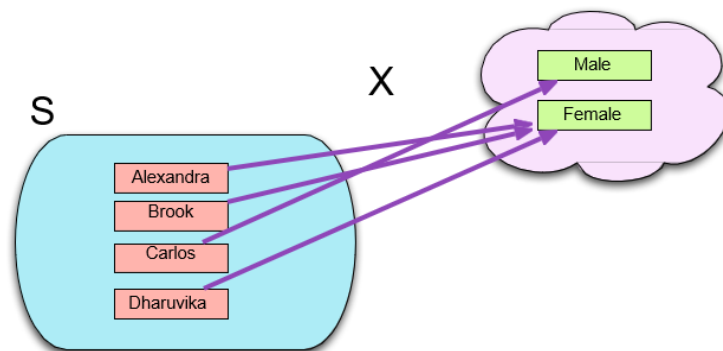
- Discrete random variable: takes a finite or countable number of distinct values.
- Continuous random variable: takes an infinite number of values within a specified range or interval.



# Distribution functions

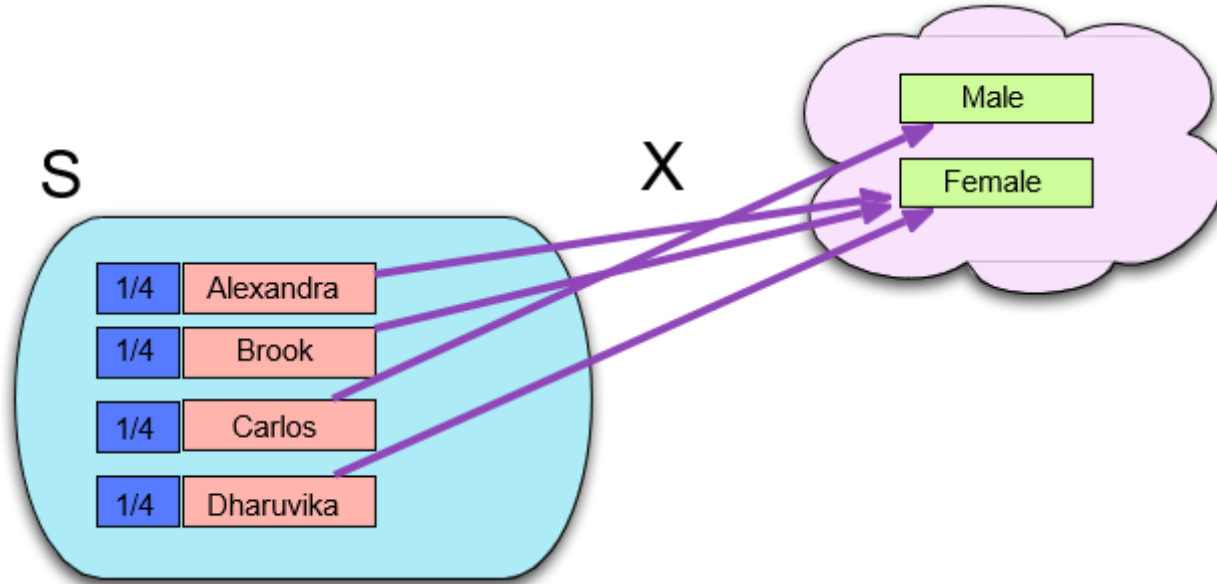
# Discrete distributions

- When a random variable is discrete, its *distribution* is characterized by the probabilities assigned to each distinct value.
- The probability that the random variable takes a particular value comes from the probability associated with the set of individual outcomes that have that value.
  - This set is an event
- E.g.  $P(X = \text{Female})$



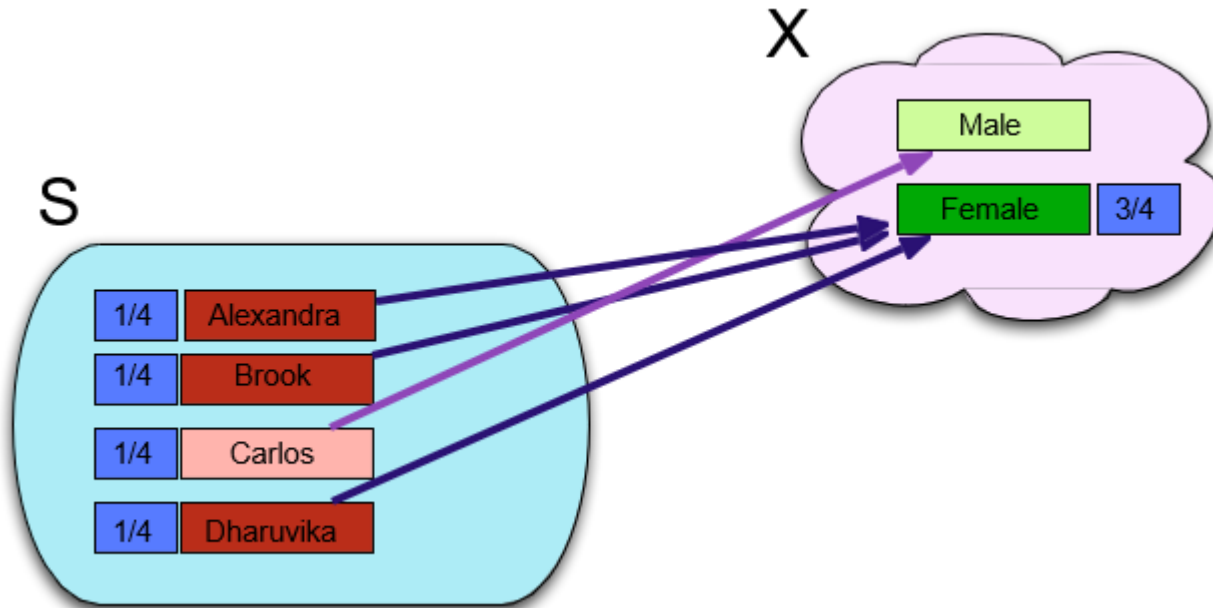
# Discrete distributions

- How to find  $P(X = \text{Female})$ ?



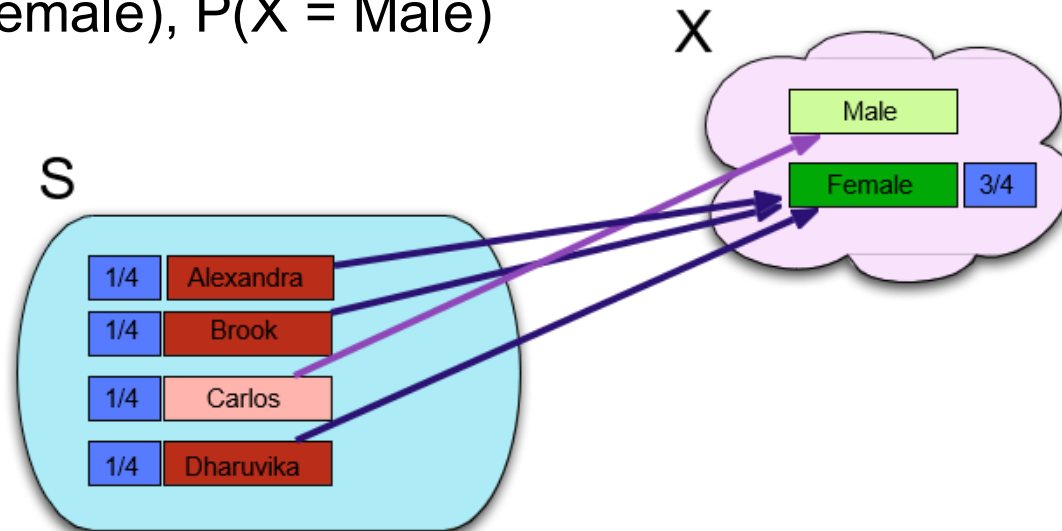
# Discrete distributions

- How to find  $P(X = \text{Female})$ ?



# Discrete distributions

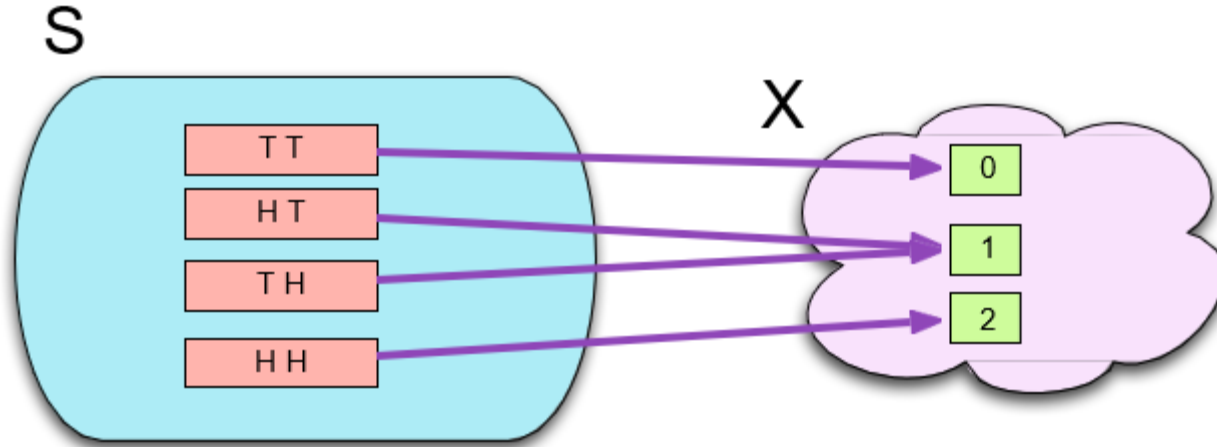
- What is the distribution of random variable  $X$ ?
  - $P(X = \text{Female})$ ,  $P(X = \text{Male})$



| $x$        | Male  | Female |
|------------|-------|--------|
| $P(X = x)$ | $1/4$ | $3/4$  |

# Discrete distributions

- What is the distribution of random variable  $X$ ?



| $x$        | 0             | 1             | 2             |
|------------|---------------|---------------|---------------|
| $P(X = x)$ | $\frac{1}{4}$ | $\frac{1}{2}$ | $\frac{1}{4}$ |

# Properties of Discrete Distributions

- We can write  $P(X = x)$  to mean “The probability that the random variable  $X$  takes the value  $x$ ”.
- What must be true of these probabilities?

## Properties of Discrete Distributions

1. Each  $P(X = x)$  is a probability, so must be between 0 and 1.
2. The  $P(X = x)$  must sum to 1 over all possible  $x$  values.

# Probability Mass function (PMF)

## The Probability Mass Function

A discrete random variable,  $X$ , can be characterized by its **probability mass function**,  $f$  (might sometimes write  $f_X$  if it's not clear from context which random variable we're talking about).

The PMF takes in values of the variable, and returns probabilities:

$f(x)$  is *defined* to be  $P(X = x)$

# PMF is a table

- Think of the PMF as a lookup table.

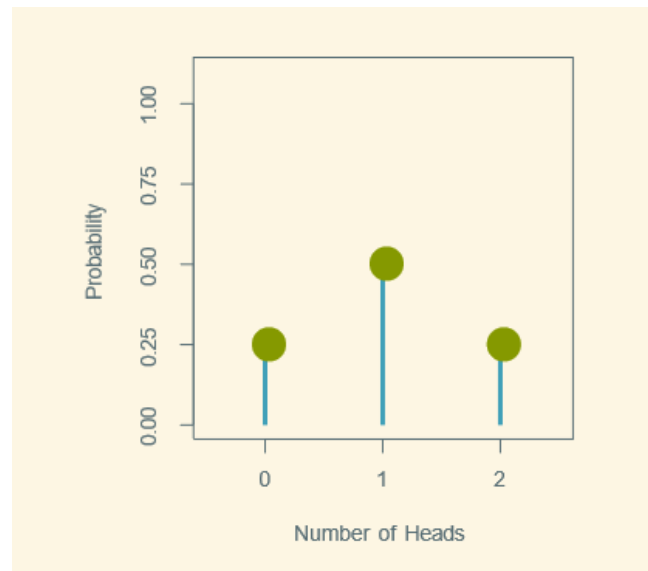
| $x$        | Male  | Female |
|------------|-------|--------|
| $P(X = x)$ | $1/4$ | $3/4$  |

- Best way to think of discrete random variables: they take various values, and each value has a certain probability of happening.

# Visualizing discrete distributions: spike plot

Flip two coins at the same time, probability distribution of number of heads:

- Often use the spike plot
- Like a bar plot, but with probabilities, instead of frequencies or proportions, on the y-axis.



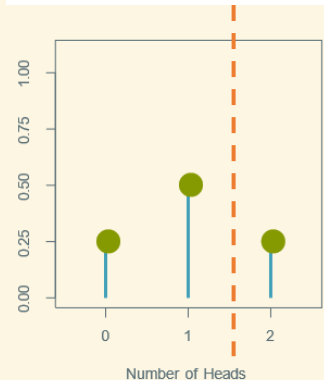
# The cumulative distribution function (CDF)

- Often, we are interested in the probability of falling in some range of values.
- We can use the cumulative distribution function (CDF), which gives the “accumulated probability” up to a particular value.

## The Cumulative Distribution Function

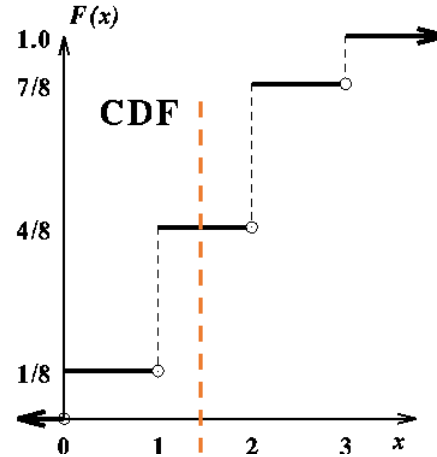
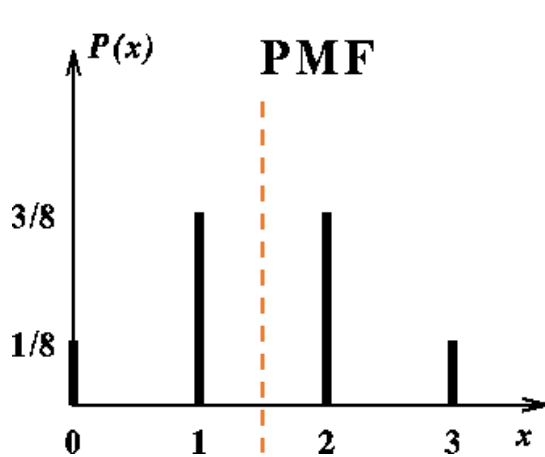
A random variable,  $X$ , can be characterized by its **cumulative distribution function**,  $F$  (or sometimes  $F_X$  if we need to be explicit), which takes values and returns *cumulative* probabilities:

$F(x)$  is defined to be  $P(X \leq x)$



# Relating PMF to CDF

- How can we calculate  $F(x)$  from the PMF table  $f$ ?
  - Add up all the probabilities up to and including  $f(x)$ .
  - What is the value of  $F(-0.1)$  (i.e.,  $P(X \leq -0.1)$ )?  $F(1.5)$ ?



- For discrete random variables,  $F(x)$  *jumps* at locations with nonzero probability mass

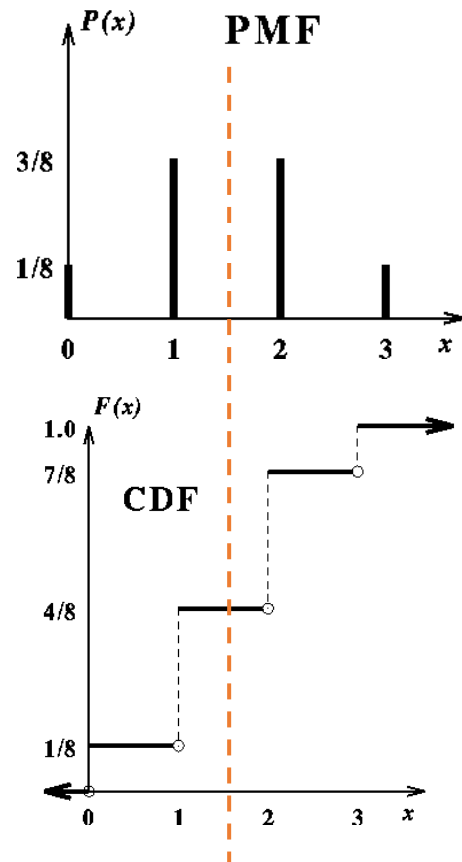
# Relating PMF to CDF

- So the PMF of  $X$  is:

$$f(x) = \begin{cases} 1/8, & x = 0 \\ 3/8, & x = 1 \\ 3/8, & x = 2 \\ 1/8, & x = 3 \end{cases}$$

- We can write the CDF of  $X$  ( $P(X \leq x)$ ):

$$F(x) = \begin{cases} 0, & x < 0 \\ \frac{1}{8}, & 0 \leq x < 1 \\ \frac{4}{8}, & 1 \leq x < 2 \\ \frac{7}{8}, & 2 \leq x < 3 \\ 1, & x \geq 3 \end{cases}$$



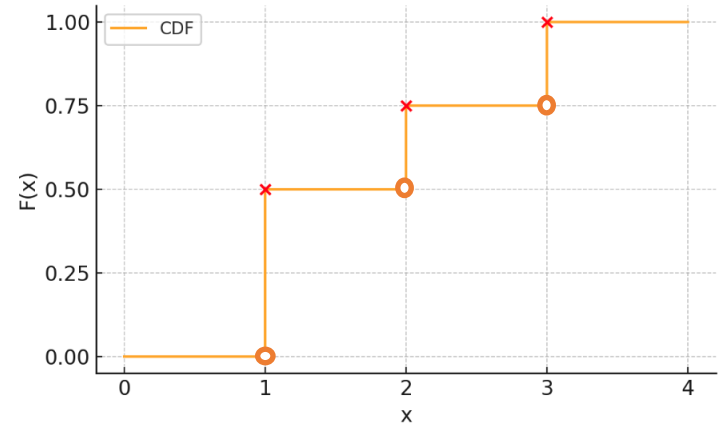
# In-class activity

- Given by the PMF of  $X$ , find the CDF of  $X$ .

$$f(x) = \begin{cases} 1/2, & x = 1 \\ 1/4, & x = 2 \\ 1/4, & x = 3 \end{cases}$$

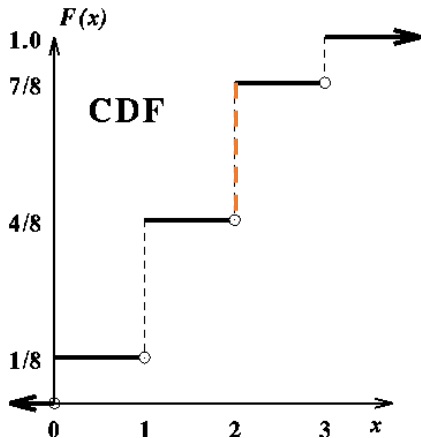
- Answer:

$$F(x) = \begin{cases} 0, & x < 1 \\ \frac{1}{2}, & 1 \leq x < 2 \\ \frac{3}{4}, & 2 \leq x < 3 \\ 1, & x \geq 3 \end{cases}$$



# Relating CDF to PMF

- How could we find  $f(x)$  from a cumulative distribution function  $F$ ? e.g.,  $f(2)$ ?

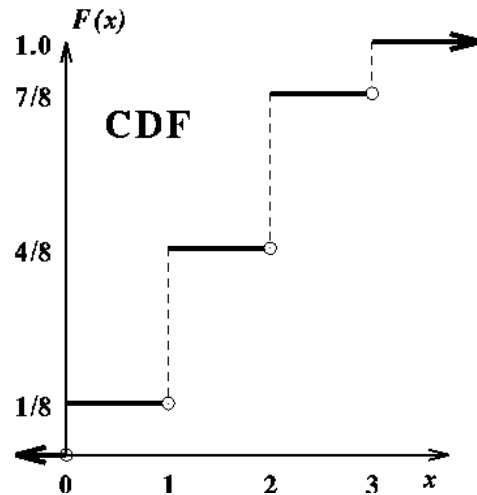


- Focus on “jumps”:  $f(x) = F(x) - F(\text{jump just below } x)$ 
  - $f(2) = F(2) - F(1) = \frac{7}{8} - \frac{4}{8} = \frac{3}{8}$
  - $f(2.1) = F(2.1) - F(2) = \frac{7}{8} - \frac{7}{8} = 0$
  - $f(1.5) = F(1.5) - F(1) = \frac{4}{8} - \frac{4}{8} = 0$

# Exercise: using CDF and PMF

Given the CDF  $F$ :

- How to calculate  $P(X > x)$ ?
  - $P(X > x) = 1 - P(X \leq x) = 1 - F(x)$
- How about  $P(X \geq x)$ ?
  - $P(X \geq x) = 1 - P(X < x) = 1 - (P(X \leq x) - P(X=x))$
  - $1 - F(x) + f(x)$
  - $f(x)$  can be 0 or nonzero, depending on whether  $x$  is a jump



# Exercise: using CDF and PMF

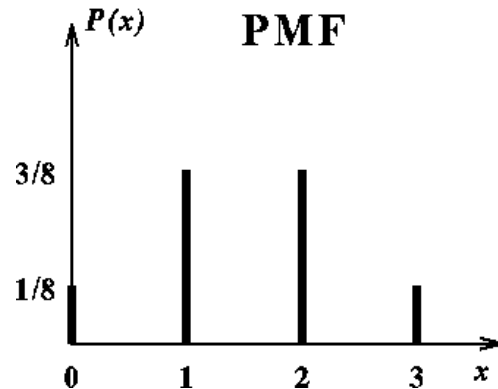
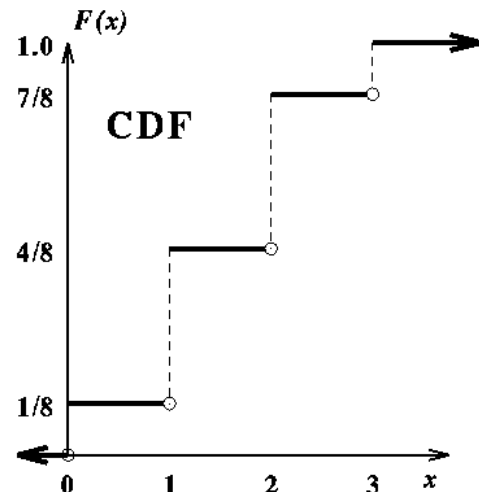
- What is  $P(X \geq 2)$ ?
  - $P(X \geq x) = 1 - F(x) + f(x)$
  - $f(x)$  can be 0 or nonzero, depending on whether  $x$  is a jump

Using the formula:

$$\bullet \quad P(X \geq 2) = 1 - F(2) + f(2) = 1 - \frac{7}{8} + \frac{3}{8} = \frac{1}{2}$$

Another way:

$$\bullet \quad P(X \geq 2) = P(X = 2) + P(X = 3) = \frac{3}{8} + \frac{1}{8} = \frac{1}{2}$$



# Exercise: using CDF and PMF

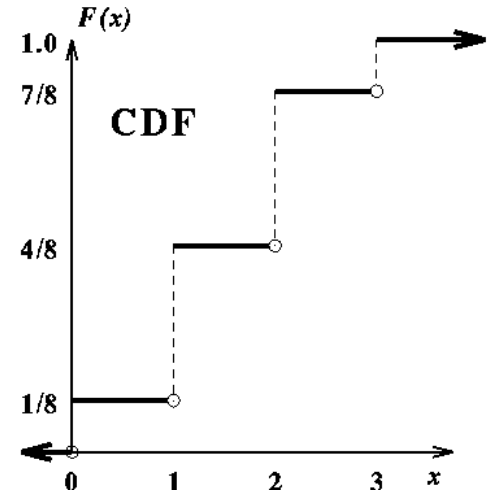
Given the CDF  $F$ :

- How to calculate  $P(a < X \leq b)$ ?

$$= P(X \leq b) - P(X \leq a)$$

$$= F(b) - F(a)$$

- How to calculate  $P(a < X < b)$ ?
  - (I'll leave this to you as an exercise..)



# Transformations of random variables

- If  $X$  is a random variable, then  $X + 5, 3X, X^2, \dots$ , are all random variables
- Given any transformation function  $f$ ,  $f(X)$  is a random variable
- How to find the PMF of  $f(X)$  based on that of  $X$ ?
  - First, find all values  $f(X)$  can take
  - For each value  $c$ , try to find  $P(f(X) = c)$

# Examples

- Suppose  $X$  has PMF

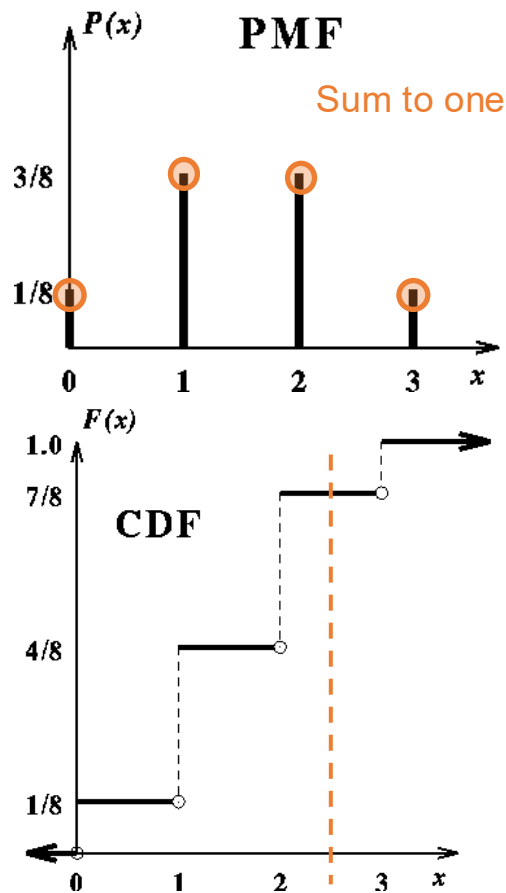
| $x$        | 1   | -1  |
|------------|-----|-----|
| $P(X = x)$ | 0.5 | 0.5 |

- What is the PMF of  $Y = X + 5$ ?
  - $Y$  can take values 6 and 4
  - $P(Y = 6) = P(X = 1) = 0.5$
  - $P(Y = 4) = P(X = -1) = 0.5$

| $y$        | 6   | 4   |
|------------|-----|-----|
| $P(Y = y)$ | 0.5 | 0.5 |

# Recap: RV, PMF and CDF

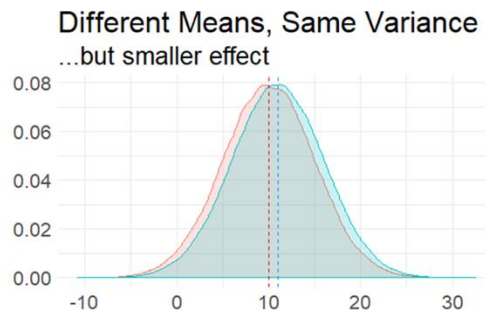
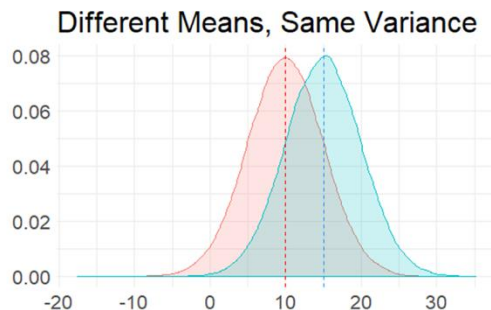
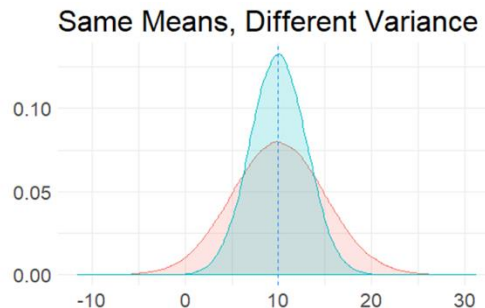
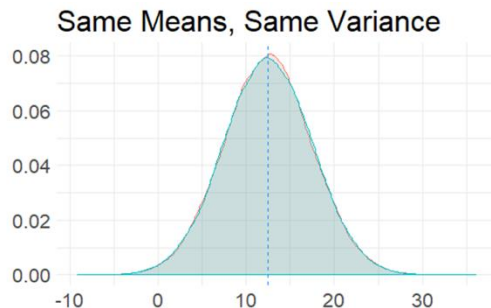
- RV: connects all outcomes to a property of interest
- A RV has a distribution, which assign a probability to each distinct value  $X$  can take
- For discrete RV  $X$ :
  - PMF:  $f(x)$  defined as  $P(X = x)$
  - CDF:  $F(x)$  defined as  $P(X \leq x)$
- Derive CDF from PMF, and vice versa
  - $f(x) = F(x) - F(\text{jump just below } x)$
  - $F(x)$ : the total of all jumps (PMF values) at points less than or equal to  $x$
- PMF of  $f(X)$ 
  - First, find all values  $f(X)$  can take
  - For each value  $c$ , try to find  $P(f(X) = c)$



# Mean and Variance

# Summarizing random variables

- It is useful to characterize the *center* and *spread* of a probability distribution
  - “what value do we expect to occur?”, and
  - “how confident are we in our prediction?”



# Mean (aka expectation, expected value)

- The mean of a random variable  $X$  is also called its *expected value*. Usually written as  $\mu$  or  $E[X]$ .
- As with a sample mean, it represents an average over the possible values; and the average is *weighted by the probabilities*.
  - $(2 + 2 + 1 + 5)/4 = 2.5$
  - $2 * \frac{1}{2} + 1 * \frac{1}{4} + 5 * \frac{1}{4} = 2.5$
- Makes sense if you were to repeat the random process many times, the average of the observed values of  $X$  would approach  $E[X]$ . It doesn't mean this value will be observed directly—it's a weighted average.

# Example: expected winnings at Roulette

- 38 outcomes (18 red, 18 black, 2 green: 0, 00) equally likely
- Suppose we bet on black. Define  $X$  which takes the value 1(\$ for outcomes where we win, and  $-1(\text{\$})$  for outcomes where we lose.
- Its probability mass function is given by

|            |       |       |
|------------|-------|-------|
| $x$        | -1    | 1     |
| $P(X = x)$ | 20/38 | 18/38 |



# Example: expected winnings at Roulette

- X's PMF is

|            |       |       |
|------------|-------|-------|
| $x$        | -1    | 1     |
| $P(X = x)$ | 20/38 | 18/38 |

- Its expected value is

$$\begin{aligned}\mu &= -1 \times P(X = -1) + 1 \times P(X = 1) \\ &= -\frac{2}{38}\end{aligned}$$

- expected value : if I play this game thousands of times, what is my average profit/loss per spin?

# Expectation formula

- Given RV  $X$  and its PMF, how to find  $E[X + 5]$ ,  $E[3X]$ , etc?
- Idea 1: find the PMF of the transformed RV and use the definition of expectation
- Idea 2: use the following fact:

## Expectation formula

$$E[f(X)] = \sum_x f(x) \cdot P(X = x)$$

# Variance

- The variance, written  $\sigma^2$  or  $\text{Var}(X)$  or  $E[(X - \mu)^2]$  is the “expected squared deviation” from the mean.
- It is a weighted average of the squared deviations corresponding to the individual values.

## Variance of a Discrete Random Variable

$$\sigma^2 \text{ (aka } \text{Var}(X), \text{ aka } E((X - \mu)^2)) = \sum_x (x - \mu)^2 P(X = x)$$

- $E[(X - \mu)^2]$  – expectation of  $(X - \mu)^2$ , another RV

# Standard deviation

- Just as with a sample, the standard deviation,  $\sigma$ , is the square root of the variance.
- E.g. in the roulette example,  $\sigma = \sqrt{0.997} \approx 0.998$ 
  - In one spin, the “typical” variation of our balance is 0.998

# Properties of expectation

- What will happen to the roulette game if we bet \$2 instead of \$1?
- The new PMF becomes
- The new expected winnings are then

|            |       |       |
|------------|-------|-------|
| $x$        | -2    | 2     |
| $P(X = x)$ | 20/38 | 18/38 |

$$\begin{aligned}\mu &= -2 \times P(X = -2) + 2 \times P(X = 2) \\ &= -\frac{4}{38}\end{aligned}$$

- What's the relationship between this value and the old expected value?
  - Doubling the individual values (w/o changing probs) doubles the expected value

# Properties of expectation

- This works in general: if we change the values of a random variable by multiplying by a constant, the expectation gets multiplied by a constant.
- To see this, recall the expectation formula:

$$E[f(X)] = \sum_x f(x) \cdot P(X = x)$$

$$E[aX] = \sum_x ax P(X = x) = a \sum_x x P(X = x) = aE[X]$$

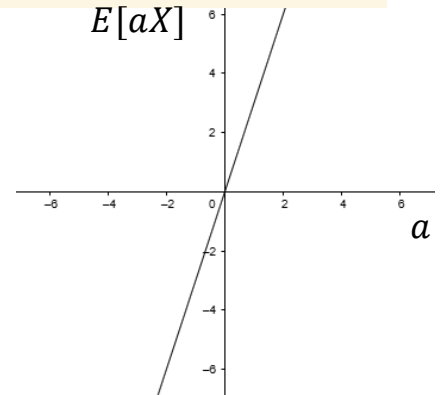
# Properties of expectation

## Property of Expectation

Multiplying a random variable by a constant scales the expected value by the same constant:

$$E(aX) = aE(X)$$

- Sometimes called “linearity of expectation”



# Properties of Variance

- What will happen to the variance if we multiply every value of a random variable by a constant  $a$ ?
- This is as if we increase our bet in the roulette game

|            |       |       |
|------------|-------|-------|
| $x$        | -2    | 2     |
| $P(X = x)$ | 20/38 | 18/38 |

- Variance = expected *squared* deviation
- All squared deviations are scaled by  $a^2$ , making variance also scaled by  $a^2$

# Properties of Variance

## Property of Variance

If the values of a random variable are multiplied by a constant,  $a$ , then the variance gets multiplied by  $a^2$ .

- In other words,  $\text{Var}(aX) = a^2 \text{Var}(X)$
- How would standard deviation change accordingly?
  - scaled by  $|a|$  (!)

# Example Discrete Random Variables

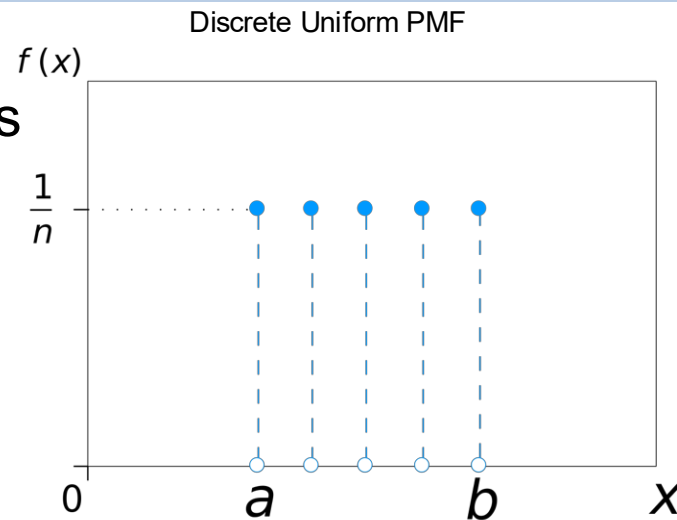
# Uniform distribution over a set

More generally, consider  $S = \{v_1, v_2, \dots, v_N\}$ ;  $X$  is drawn from the uniform distribution of  $S$ , then

$$P(X = k) = \begin{cases} \frac{1}{N} & \text{if } k \in \{v_1, v_2, \dots, v_N\} \\ 0 & \text{otherwise} \end{cases}$$

We denote this by  $X \sim \text{Uniform}(S)$

- Selecting a student from a class
- Drawing a card from a shuffled deck
- Choosing a letter from the alphabet



To generate a sample from a uniform discrete distribution,

```
random.choice(a, size=None, replace=True, p=None)
```

Generates a random sample from a given 1-D array

```
numpy.random.choice([2,5,6])
```

Example output: 2

# Binomial distribution

- Suppose we perform  $n$  repeated independent trials, each with success probability  $p$ , what is the distribution of the number of successes  $X$ ?

- What values can  $X$  take?

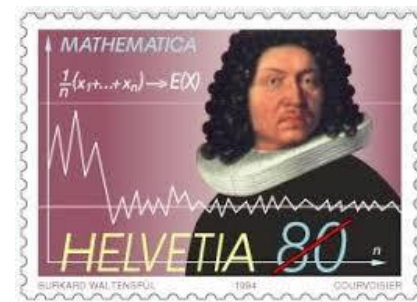
$$m = 0, 1, \dots, n$$

- We have seen that  $P(X = m) =$

$$\binom{n}{m} \cdot p^m (1 - p)^{n-m}$$

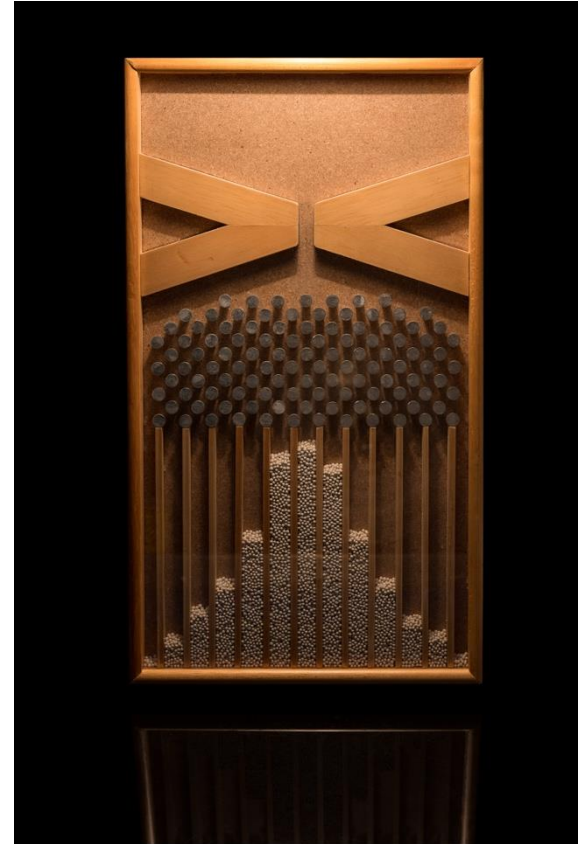
- In this case,  $X$  is said to be drawn from a *binomial distribution*, denoted by

$$X \sim \text{Bin}(n, p)$$



# Galton Boards

- Illustration of binomial distribution
- Bead has 10 chances hitting pegs (10 rows of pegs)
- each time a peg is hit, bead randomly bounces to the left or the right with equal probabilities
- Number of times it bounces to the left:  
 $X \sim \text{Bin}(10, 0.5)$

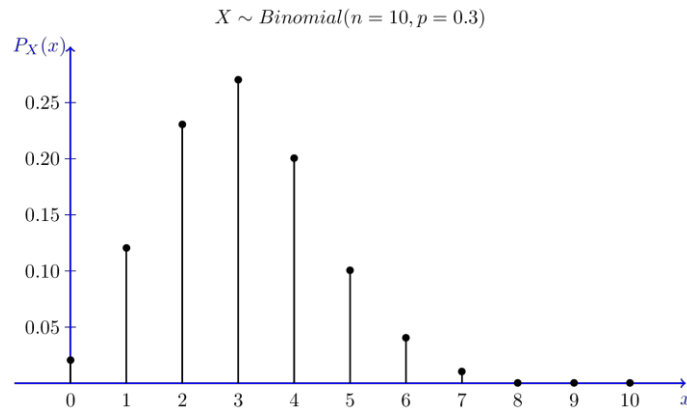


# Binomial distribution

- $X \sim \text{Bin}(n, p)$
- $X$ 's PMF is “Bell-shaped”

Facts:

- $E[X] = E[n \cdot X_i] = n \cdot E[X_i] = np$
- $\text{Var}[X] = np(1 - p)$ 
  - Small when  $p$  is close to 0 or 1



# Bernoulli distribution

- What does  $X \sim \text{Bin}(1, p)$  mean?

| $x$        | 0     | 1   |
|------------|-------|-----|
| $P(X = x)$ | $1-p$ | $p$ |

- This is called the Bernoulli distribution with parameter  $p$ , abbreviated as Bernoulli( $p$ )
- $E[X] = 0 \cdot (1 - p) + 1 \cdot p = p$

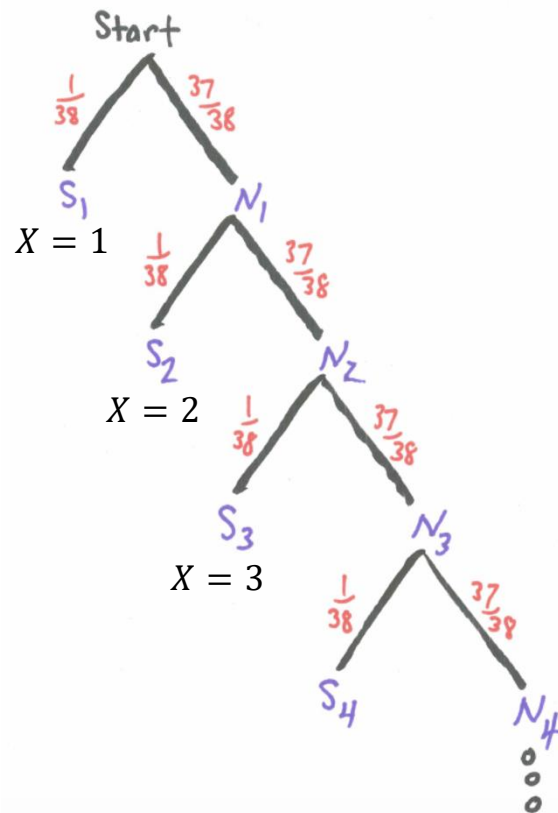


# Geometric distribution

- Suppose we perform repeated independent trials with success probability  $p$ . What is the distribution of  $X$ , the number of trials needed to get a success? (related to Q4 in HW3)
- Applications:
  - Call center: # calls before encountering first dissatisfied customer
  - Basketball: # shots before scoring the first
  - Networking: # attempts before a successful transmission
  - Gambling: # plays before first win

# Geometric distribution

- How to find  $P(X = x)$ ?
- Let's draw a probability tree..
- Example:  $p = \frac{1}{38}$  (roulette)
- $P(X = 1) = p$
- $P(X = 2) = (1 - p) p$
- $P(X = 3) = (1 - p)^2 p$
- ...



# Geometric distribution

- In conclusion,

$$P(X = x) = p (1 - p)^{x-1}$$

for  $x = 1, 2, \dots$

Fact:

- $E[X] = \frac{1}{p}$
- $\text{Var}[X] = \frac{1-p}{p^2}$ 
  - Smaller when  $p$  closes to 1

